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3.2.1 The Measurement of Macroeconomic Performance

The Four Main Macroeconomic Objectives

The UK government pursues four primary macroeconomic objectives. These form the framework for evaluating economic performance:

Objective	Target/Measure	Why It Matters
1. Economic growth	Sustained increase in real GDP (trend growth ~2-2.5% per year)	Rising GDP means more goods and services produced → higher living standards, more jobs, higher tax revenue
2. Low and stable inflation	CPI target of **2%** (set by government, implemented by Bank of England)	High/volatile inflation erodes purchasing power, creates uncertainty, distorts decision-making, and hurts those on fixed incomes
3. Low unemployment	Full employment (minimising involuntary unemployment, not zero unemployment)	Unemployment wastes resources (economy inside PPF), causes poverty, deskilling, social problems, and reduces tax revenue
4. Balance of payments equilibrium	Avoiding persistent, unsustainable current account deficits	Large deficits may indicate lack of competitiveness and require foreign borrowing to finance

Additional Objectives

- **Fair distribution of income and wealth** — reducing inequality (measured by the Gini coefficient)
- **Environmental sustainability** — growth that does not deplete resources or degrade the environment for future generations
- **Sound public finances** — manageable government budget deficit and national debt
- **Exchange rate stability** — avoiding excessive volatility that harms trade and investment

Conflicts Between Objectives

Achieving all objectives simultaneously is extremely difficult because many conflict with each other:

- **Growth vs inflation:** Rapid growth can cause demand-pull inflation (too much spending chasing too few goods)
- **Low unemployment vs inflation:** The Phillips Curve suggests a short-run trade-off — reducing unemployment may increase inflationary pressure as wages rise
- **Growth vs balance of payments:** Economic growth increases imports (consumers have more income to spend on foreign goods), worsening the current account
- **Growth vs environment:** Higher production can increase pollution and resource depletion
- **Low inflation vs low unemployment:** Tight monetary policy (high interest rates) reduces inflation but may increase unemployment

Real-World Example: The UK in 2022-23

In 2022-23, the UK faced acute conflicts between objectives. Inflation surged to 11.1% (October 2022) — the highest in 40 years — driven by energy prices and post-COVID supply chains. The Bank of England raised interest rates from 0.1% to 5.25% to combat inflation. But these rate rises slowed economic growth (the UK narrowly avoided recession) and risked increasing unemployment. Mortgage holders faced sharply higher payments, reducing disposable income. The government had to choose between fighting inflation (higher rates) and supporting growth (lower rates) — it couldn't do both simultaneously.

Gross Domestic Product (GDP)

Gross Domestic Product (GDP) is the total monetary value of all *final* goods and services produced within a country's borders in a given time period (usually a year or quarter). It is the most

widely used measure of economic output and activity.

Three Methods of Measuring GDP

All three methods should theoretically give the same result, because every pound spent (expenditure) must be earned by someone (income) and must have been generated by production (output). This is the **circular flow of income**.

Method	What It Measures	Formula / Components
Expenditure method	Total spending on final goods and services in the economy	**GDP = C + I + G + (X − M)** C = consumer spending I = investment (capital spending by firms) G = government spending X = exports, M = imports
Income method	Total income earned by all factors of production	Wages + Rent + Interest + Profit (all factor incomes generated within the country)
Output method	Total value added by every industry/sector	Sum of value added across all sectors (agriculture, manufacturing, services, etc.) Value added = output − intermediate goods (to avoid double counting)

Exam Tip: The expenditure formula **C + I + G + (X − M)** is essential — learn it thoroughly. It appears in questions about components of aggregate demand, fiscal policy, trade, and economic growth. Each component can be analysed separately.

Real GDP vs Nominal GDP

Nominal (money) GDP — GDP measured at *current* prices. Changes in nominal GDP can be caused by changes in output OR changes in prices. If prices rise 10% and output stays the same, nominal GDP rises 10% — but nothing more has actually been produced.

Real GDP — GDP adjusted for inflation, measured at *constant* (base-year) prices. This strips out the effect of price changes, showing the true change in the volume of goods and services produced.

$$\text{Real GDP} = (\text{Nominal GDP} \div \text{GDP Deflator}) \times 100$$

Real GDP is always the more meaningful measure for assessing economic performance. A country whose nominal GDP grows by 5% but has 5% inflation has experienced zero real growth — living standards haven't improved at all.

GDP Per Capita

$$\text{GDP per capita} = \text{Real GDP} \div \text{Population}$$

This measures the average output (and roughly, the average income) per person. It is a better indicator of living standards than total GDP because it accounts for population differences. China's total GDP is larger than the UK's, but its GDP per capita is much lower because it has 20 times more people.

Real-World Data: UK GDP

UK real GDP was approximately £2.27 trillion in 2023. GDP per capita was approximately £33,700. For international comparison: US GDP per capita was about $76,000, Germany's about €46,000, and India's about $2,600. These figures need to be adjusted for Purchasing Power Parity (PPP) to be meaningful — $1 buys much more in India than in the US, so a raw comparison overstates the gap in living standards.

Economic Growth

Economic growth is an increase in real GDP over time. It can be measured as:
Short-run growth: Actual growth — the % change in real GDP from one period to the next. This measures how much more the economy is producing.
Long-run growth: An increase in the economy's productive potential (capacity) — shown by an outward shift of the PPF or the LRAS curve.

UK trend growth rate is approximately 2-2.5% per year. Growth above this risks inflation; persistently below this suggests underlying structural problems.

The Economic (Business/Trade) Cycle

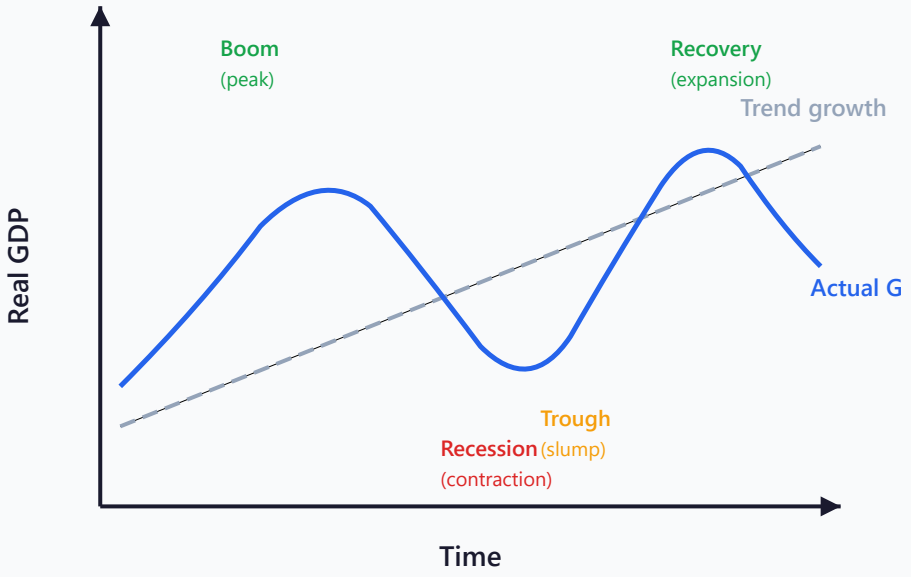

Figure 1: The economic cycle — actual GDP fluctuates around the long-run trend growth path

Phases of the Economic Cycle

Phase	Characteristics
Boom (peak)	High/rising real GDP, low unemployment, rising wages, consumer confidence high, possible inflation as economy overheats, current account may worsen (high imports)
Recession (contraction)	Falling real GDP (technically: two consecutive quarters of negative growth), rising unemployment, falling consumer confidence, business investment falls, tax revenue falls, government spending on benefits rises
Trough (slump)	The lowest point of the cycle. GDP is at its furthest below trend. Unemployment at its highest. Prices may be falling (deflation).

Recovery (expansion)	GDP begins to rise again, unemployment starts falling, confidence returns, investment picks up, spending increases

Real-World Example: COVID-19 Recession (2020)

The UK experienced its deepest recession on record in 2020. Real GDP fell by 9.7% — the largest annual decline since records began in 1948 (far worse than the 2008 financial crisis, which saw a 4.2% decline). Unemployment rose from 3.8% to 5.2%. However, the recovery was also rapid: GDP grew by 7.6% in 2021 as restrictions eased and pent-up demand was released. This V-shaped pattern was unusual — most recessions have slower recoveries.

Limitations of GDP as a Measure of Living Standards

While GDP is the most commonly used measure of economic performance, it has significant limitations as a measure of wellbeing and living standards:

Limitation	Explanation	Example
Income distribution	GDP per capita is an average — it hides inequality. A country could have high average GDP but extreme poverty alongside extreme wealth.	Qatar has the world's highest GDP per capita, but migrant workers (80%+ of population) live in poor conditions
Shadow/informal economy	Unreported economic activity (cash-in-hand work, illegal markets) is not counted in GDP. This varies hugely between countries.	Italy's shadow economy is estimated at 20%+ of GDP; the UK's is about 10%
Non-marketed output	Household work (cooking, cleaning, childcare), voluntary work, and DIY are not counted despite having real economic value.	If a parent cares for their child at home: not counted. If they pay a nursery: counted in GDP.

Quality improvements	GDP doesn't fully capture quality improvements. A smartphone today is vastly better than one from 2010, but GDP may not reflect this quality gain.	Many digital services are free (Google, Wikipedia) — they improve lives but add nothing to GDP
Environmental degradation	GDP counts the production that causes pollution but doesn't deduct the environmental damage. A country could boost GDP by destroying its forests.	China's rapid GDP growth came with severe air pollution — GDP didn't account for the health costs
Leisure time	Higher GDP may come at the cost of longer working hours and less leisure. People working 60-hour weeks have a higher GDP contribution but may have lower wellbeing.	South Korea has high GDP per capita but extremely long working hours compared to Germany
Composition of output	GDP counts all spending equally. Military spending, pollution clean-up, and rebuilding after natural disasters all increase GDP but don't improve welfare.	After a hurricane, GDP rises as rebuilding occurs — but people are clearly worse off
PPP comparisons	International comparisons using market exchange rates are misleading because the same dollar buys different amounts in different countries.	$1 buys much more in Vietnam than in Switzerland — PPP adjustment is essential

Alternative Measures of Living Standards

Measure	What It Includes	Advantage Over GDP
Human Development Index (HDI)	Life expectancy, education (years of schooling), and GNI per capita (PPP)	Broader than GDP alone — captures health and education outcomes

Genuine Progress Indicator (GPI)	Starts with personal consumption, then adds/subtracts for inequality, environmental damage, crime, leisure, household work, etc.	Accounts for environmental and social costs that GDP ignores
Inequality-adjusted HDI	HDI adjusted downward for inequality within a country	Penalises countries with high average HDI but extreme inequality
Happy Planet Index	Wellbeing × life expectancy ÷ ecological footprint	Measures sustainable wellbeing — how efficiently resources are converted into long, happy lives

Exam Tip: When evaluating GDP data, always cite at least 2-3 specific limitations relevant to the question. Then suggest an alternative measure that would be more appropriate. This shows the examiner you can think critically about data, not just recite numbers.

Inflation

Inflation is a sustained increase in the general (average) price level over time. It means the purchasing power of money falls — each pound buys fewer goods and services.

Deflation is a sustained decrease in the general price level. Prices are falling on average.

Disinflation is a fall in the *rate* of inflation — prices are still rising, but more slowly than before. (E.g., inflation falling from 10% to 5% is disinflation, not deflation.)

Measuring Inflation: CPI and RPI

Feature	CPI (Consumer Prices Index)	RPI (Retail Prices Index)

Used for	Official UK inflation target (2%), Bank of England monetary policy	Some pension uprating, student loan interest, some contracts
Includes mortgage interest payments?	No	Yes
Includes council tax?	No	Yes
Mathematical formula	Geometric mean (tends to give a lower figure)	Arithmetic mean (tends to give a higher figure)
Coverage	Wider — covers virtually all UK households	Excludes top 4% earners and pensioners mainly dependent on state benefits
International comparability	Yes — harmonised across EU/OECD	UK-specific only
Typical difference	RPI is usually about 0.5-1.0 percentage points higher than CPI	

CPIH is a newer measure that adds owner-occupier housing costs (OOH) to CPI. It has been the ONS's preferred measure since 2017.

How CPI Is Calculated

1. **Select a representative "basket of goods"** — around 700 items that represent typical household spending. The basket is updated annually to reflect changing patterns (e.g., streaming services added, DVD players removed).
2. **Collect prices** — ONS price collectors record prices of these items from shops, websites, and service providers across the UK each month (around 180,000 prices).
3. **Apply weights** — each item is given a weight based on its share of average household spending. Food has a larger weight than cinema tickets because households spend more on food.
4. **Calculate the index** — combine weighted price changes into a single number relative to a base year (base year = 100). Compare to the previous year to get the inflation rate.

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Limitations of CPI

- **Not representative for individuals:** CPI measures average household spending. A pensioner, a student, and a family of four have very different spending patterns and experience different personal inflation rates.
- **Quality adjustments:** When a product improves in quality (e.g., faster smartphones), price increases partly reflect better quality, not just inflation. Adjusting for this is subjective and imprecise.
- **New products lag:** New goods (e.g., electric scooters, air fryers) are not immediately included in the basket. Their prices may fall rapidly initially, which CPI misses.
- **Excludes housing costs:** CPI (unlike CPIH) does not include mortgage interest payments — a major expense for many households.
- **Substitution bias:** When the price of one good rises, consumers switch to cheaper substitutes. CPI may overstate inflation because the basket doesn't update fast enough to reflect this switching.
- **Different inflation for different groups:** Low-income households spend a higher proportion on food and energy. When food and energy prices spike, they face higher effective inflation than the headline CPI suggests.

Real-World Example: The 2022-23 Cost-of-Living Crisis

In 2022, UK CPI inflation hit 11.1% — the highest since 1981. But the experience varied hugely by household. Energy bills rose by 54% in April 2022 and another 27% in October. Food prices rose by over 16%. Low-income households, who spend a higher share of income on energy and food, experienced effective inflation rates of 15%+, while higher-income households (more spending on services, which rose less) experienced perhaps 8-9%. The single CPI figure masked this inequality. This shows why CPI is a limited measure of how inflation affects real people.

Unemployment

Unemployment occurs when people who are willing and able to work, and are actively seeking employment, cannot find a job. It represents a waste of human resources — the economy is operating inside its PPF.

Two Ways of Measuring Unemployment

	Claimant Count	ILO Labour Force Survey (LFS)
Definition	Number of people claiming unemployment-related benefits (Jobseeker's Allowance or Universal Credit job-seeking element)	Number of people without a job who have actively sought work in the last 4 weeks AND are available to start within 2 weeks
Data source	Administrative records (benefit claims)	Survey of 40,000 households per quarter
Advantages	Cheap, accurate count, available monthly, no sampling error	Internationally comparable (ILO standard), covers people not eligible for benefits (e.g., recent graduates, returning carers)
Disadvantages	Excludes those ineligible or not claiming benefits, sensitive to policy changes (tightening benefit rules reduces count without reducing actual unemployment)	Based on a sample (subject to sampling error), published with a time lag, self-reported (relies on honesty)

The ILO/LFS measure is generally considered more reliable and is the official measure used for international comparisons. However, neither measure captures **underemployment** (people working fewer hours than they want) or **hidden unemployment** (discouraged workers who have stopped looking).

Types of Unemployment

Type	Definition	Cause	Real-World Example
Frictional	Short-term unemployment that occurs as workers move between jobs. They are in between positions, not structurally unemployed.	Normal labour market turnover, time taken to search for and find suitable new work, imperfect information about vacancies	A graduate taking 3 months to find their first job; someone who quits to find a better role

Structural	Long-term unemployment caused by a mismatch between workers' skills/location and the skills/locations that employers need.	Deindustrialisation, technological change, geographical immobility (workers can't move to where jobs are), occupational immobility (workers lack the right skills)	Ex-coal miners in South Wales unable to find work when mines closed; factory workers replaced by automation
Cyclical (demand-deficient)	Unemployment caused by insufficient aggregate demand in the economy during a recession.	A fall in AD → firms produce less → lay off workers → further falls in spending (multiplier effect) → more job losses	Mass layoffs during the 2008 financial crisis and the 2020 COVID recession
Seasonal	Unemployment that occurs at predictable times of year due to seasonal patterns in certain industries.	Industries with seasonal demand: tourism, agriculture, retail (post-Christmas layoffs), ski resorts	Seaside hotel workers unemployed in winter; agricultural labourers between harvests
Real wage / Classical	Unemployment caused by wages being set above the market-clearing (equilibrium) level.	National Minimum Wage/Living Wage (if set above equilibrium), trade union wage bargaining, "sticky" wages that don't fall in a downturn	Theory suggests minimum wage could price low-skilled workers out of jobs, though evidence is mixed

Costs of Unemployment

Costs to Individuals

- **Loss of income:** Reduced living standards, potential poverty, inability to service debts (mortgage arrears)

- **Deskilling:** The longer someone is unemployed, the more their skills and work habits deteriorate, making them less employable — a vicious circle
- **Health problems:** Unemployment is strongly correlated with stress, depression, anxiety, and physical health problems
- **Social exclusion:** Loss of social networks, stigma, reduced self-esteem

Costs to the Economy/Government

- **Lost output:** The economy operates inside the PPF — resources are being wasted
- **Reduced tax revenue:** Fewer workers paying income tax and national insurance
- **Increased government spending:** Higher benefit payments (JSA, Universal Credit, housing benefit)
- **Hysteresis:** Long-term unemployment can become self-perpetuating. Workers lose skills and motivation, and employers are reluctant to hire them. Even when the economy recovers, unemployment may not fall back to its pre-recession level. This is why preventing unemployment is often more important than curing it.

Costs to Society

- Higher crime rates (particularly in areas of high youth unemployment)
- Regional deprivation — unemployment is often concentrated in specific areas (former industrial towns in the North of England)
- Family breakdown and domestic problems
- Wasted human potential

Real-World Example: UK Regional Unemployment Disparities

UK unemployment is not evenly distributed. In 2023, the unemployment rate in the South East was about 3.2%, while in the North East it was about 5.1%. Former industrial areas like parts of South Wales, Teesside, and the West Midlands have experienced structural unemployment since deindustrialisation in the 1980s. Workers who lost manufacturing jobs often lacked the skills for the new service-sector jobs being created in London and the South East, and geographical immobility (housing costs, family ties) prevented them from moving. This structural mismatch persists decades later.

Balance of Payments

The Balance of Payments (BoP) is a systematic record of all economic transactions between residents of a country and the rest of the world over a given time period. It always balances overall (the sum of all three accounts equals zero, plus a statistical discrepancy).

Structure of the Balance of Payments

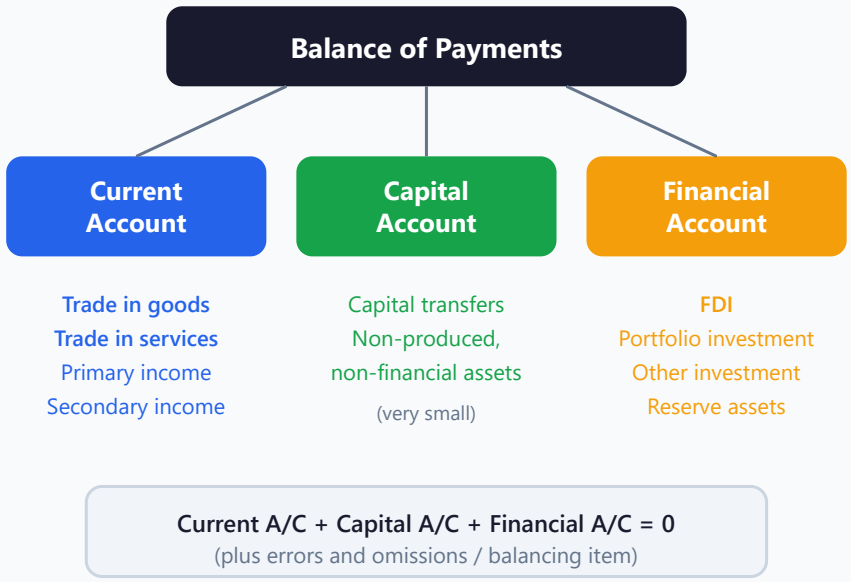

Figure 2: The three accounts of the Balance of Payments — they must sum to zero overall

The Current Account in Detail

The current account is the most frequently discussed component. It records:

Component	What It Records	UK Position
Trade in goods (visible trade)	Exports minus imports of physical goods (cars, food, oil, machinery, clothing)	**Deficit** — the UK imports far more goods than it exports (deficit of about £150-180 billion/year). The UK has deindustrialised

		and is a net importer of manufactured goods.
Trade in services (invisible trade)	Exports minus imports of services (financial services, insurance, consulting, tourism, IT, education)	**Surplus** — the UK is one of the world's largest exporters of financial and professional services. London is a global financial centre. Surplus of about £100-130 billion/year.
Primary income	Income earned from overseas investments (dividends, interest, wages) minus income paid to foreign investors in the UK	Variable — depends on relative investment returns at home and abroad
Secondary income (transfers)	Government transfers (foreign aid) and private transfers (remittances sent abroad by foreign workers in the UK)	**Deficit** — the UK is a net donor of aid and has large outward remittance flows

The UK's Persistent Current Account Deficit

The UK has run a current account deficit almost every year since the early 1980s. The overall current account deficit has been typically 3-5% of GDP in recent years.

Causes of the UK's Deficit

- **Deindustrialisation:** Decline in manufacturing means fewer goods to export
- **High domestic demand:** UK consumers have a high propensity to import (cars, electronics, food, clothing — much is imported)
- **Strong exchange rate periods:** A strong pound makes UK exports expensive and imports cheap
- **Low productivity growth:** UK firms are often less competitive than German, Japanese, or South Korean manufacturers
- **Energy imports:** The UK became a net importer of oil and gas after North Sea production peaked in the late 1990s

Is a Deficit a Problem?

Arguments that a current account deficit IS a problem:

- Must be financed by borrowing from abroad or selling UK assets to foreign investors — increasing foreign ownership of the UK economy
- May indicate a lack of international competitiveness — UK firms cannot compete on price or quality
- Persistent deficits may lead to currency depreciation — increasing import costs and inflation
- If foreign investors lose confidence, they may withdraw capital suddenly — causing a currency crisis

Arguments that a deficit is NOT necessarily a problem:

- The deficit may reflect **strong economic growth** — growing economies import more (a sign of prosperity, not weakness)
- If financed by FDI (foreign direct investment), it brings new technology, jobs, and skills to the UK
- The UK's surplus in services partly offsets the goods deficit — the economy is restructuring, not failing
- Many advanced economies run persistent deficits (US, UK, Australia) without crisis
- The key question is **sustainability** — is the deficit being financed in a healthy way?

Index Numbers

Index numbers are used to show the value of a variable relative to a base period (set at 100). They make it easy to see percentage changes over time without dealing with raw data in different units.

$$\text{Index number} = (\text{Value in current period} \div \text{Value in base period}) \times 100$$

Example: If the base year price of a basket of goods is £500 and the current year price is £540:

$$\text{Index} = (540 \div 500) \times 100 = 108 \rightarrow \text{prices have risen by } 8\%$$

Common Uses of Index Numbers

- **CPI / RPI:** Measuring inflation (base year = 100)
- **Real GDP index:** Measuring economic growth over time
- **Trade-weighted exchange rate index:** Measuring the overall strength of a currency against a basket of trading partners' currencies
- **Share price indices:** FTSE 100 (UK top 100 companies), S&P 500 (US top 500)
- **House price index:** Tracking changes in property values over time

Exam Tip: When interpreting index numbers, remember: the number itself doesn't tell you the actual value — it tells you the change relative to the base year. An index of 120 means a 20% increase from the base year. An index falling from 120 to 115 does NOT mean prices have fallen — it means the rate of increase has slowed (disinflation if we're talking about prices).

Uses and Limitations of National Income Data

Uses of National Income Data

- **Measuring economic growth:** Tracking changes in real GDP over time to assess whether the economy is expanding or contracting
- **International comparisons:** Comparing GDP per capita (at PPP) between countries to assess relative living standards
- **Government policy evaluation:** Assessing whether policies (fiscal stimulus, monetary easing) are achieving their objectives
- **Identifying trends and cycles:** Recognising whether the economy is in boom, recession, or recovery — informing forward-looking policy
- **Resource allocation:** Governments use GDP data to allocate funding between regions, public services, and infrastructure priorities
- **Business planning:** Firms use GDP forecasts to plan investment, hiring, and production decisions

Key Evaluation Points

When evaluating any macroeconomic data in an exam:

- Always distinguish between **real** and **nominal** values
- Consider the **distribution** — averages can be misleading
- Acknowledge **what the data excludes** (shadow economy, environmental damage, wellbeing)
- Note the **time period** — a single year's data may not reflect the underlying trend
- Consider **international comparisons** using PPP, not market exchange rates
- Use **alternative indicators** (HDI, Gini coefficient) to complement GDP data
- Remember that data collection is imperfect — GDP figures are revised multiple times after initial publication

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